

The recent advent of multidrug-resistant *Mycobacterium tuberculosis* (Mtb) has led to the search for novel agents to combat tuberculosis. Mtb secretes over 70 serine proteases that may be used as viable drug targets. Hydrolase important for pathogenesis 1 (Hip1) is a serine protease found in Mtb with proteolytic activity that is required for dampening the host inflammatory response. Given the role of Hip1 in Mtb virulence, analyzing Hip1 substrate specificity is a critical step in designing selective inhibitors for effective tuberculosis treatment.

Fluorogenic substrates are nonfluorescent compounds that release a fluorescent reporter group when acted upon by an enzyme. Such substrates are useful biomolecular imaging tools that allow monitoring of enzymatic activity *in vitro* and *in vivo*. A large library of peptides was screened against Hip1, and enzymatic cleavage was analyzed using mass spectrometry to facilitate the design of fluorogenic substrates (Figure 1). The cleavage reaction has a large, negative ΔG° under the conditions used.

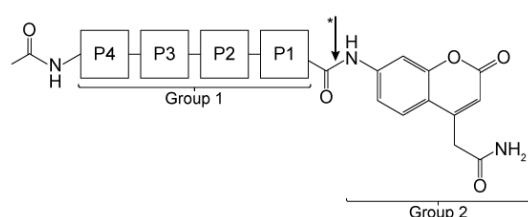


Figure 1 Schematic design of Hip1 fluorogenic substrates. P1, P2, P3, and P4 in Group 1 represent the positions of variable amino acids in the peptide. Group 2 depicts the fluorescent compound released on Hip1 cleavage.

Researchers reported Michaelis–Menten kinetic parameters for substrates from two peptide families: the WKLL substrates and the CSL substrates.

Table 1 Kinetic Parameters of Fluorogenic Substrates for Mtb Hip1

Substrate	k_{cat} (s^{-1})	K_m (M)	$\frac{k_{\text{cat}}}{K_m}$ ($\text{s}^{-1}\text{M}^{-1}$)
WKLL-ACC	1	8.3×10^{-5}	1.2×10^4
CSL-173	15	8.8×10^{-7}	1.7×10^7
CSL-174	6	2.4×10^{-6}	2.5×10^6
CSL-175	12	1.6×10^{-6}	7.5×10^6
CSL-176	3	1.5×10^{-7}	2×10^7

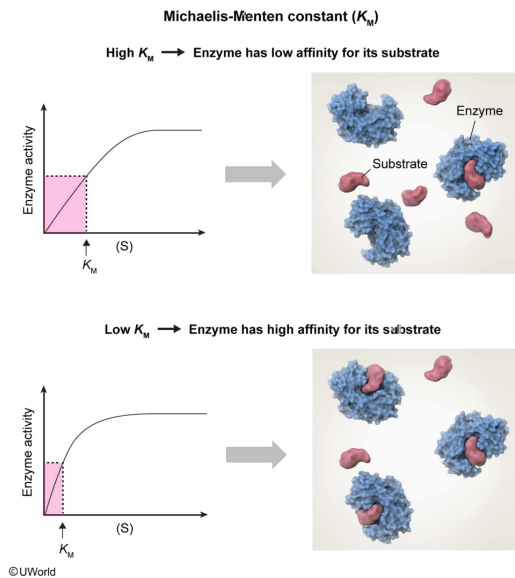
Assuming that product formation is the rate-limiting step for all substrates, which of the following statements regarding the formation of the Hip1 enzyme–substrate (ES) complex is correct?

- ☐ A. WKLL-ACC forms a stronger ES complex than CSL-175. [15%]
- ☒ B. CSL-174 forms a stronger ES complex than WKLL-ACC. [74%]
- ☐ C. CSL-176 forms a weaker ES complex than CSL-175. [7%]
- ☐ D. CSL-175 forms a weaker ES complex than CSL-174. [3%]

Correct

73% answered correctly

Explanation:



The **Michaelis–Menten constant K_m** is the substrate concentration at which **half of the reaction's maximum velocity** is achieved. It depends on both the rate of substrate binding to the enzyme (ES complex formation) and the rate at which the bound substrate is converted to product (product formation). Because the question indicates that the rate of product formation is significantly slower than the rate of ES complex formation (ie, product formation is rate-limiting), k_{cat} must be small compared to the binding constant K_m . Therefore, in this case K_m essentially measures the enzyme's **affinity for a substrate**, or the tendency of an enzyme and a substrate to bind and form a complex.

When product formation is rate-limiting, K_m can also be thought of as the substrate concentration at which half of the enzymes in solution are bound. A **small K_m** indicates a **strong ES complex** because only a small amount of substrate is required to achieve a large amount of complex formation. On the other hand, a **large K_m** indicates a **weak ES complex** that does not readily form. According to Table 1, CSL-174 has a smaller K_m for Hip1 than WKLL-ACC does, so CSL-174 forms a stronger ES complex.

(Choice A) CSL-175 has a smaller K_m than WKLL-ACC, which means that WKLL-ACC forms a weaker ES complex. In fact, WKLL-ACC has the highest K_m of all, and therefore forms the weakest complex.

(Choice C) CSL-176 has the smallest K_m of the peptides tested, and therefore forms the strongest ES complex.

(Choice D) CSL-175 has a smaller K_m than CSL-174, so it forms a stronger complex.

Educational objective:

The Michaelis-Menten constant K_m is an approximate measure of an enzyme's affinity for a substrate or of the strength of the enzyme-substrate complex. A low K_m represents a high affinity and a strong complex whereas a high K_m indicates a low affinity and a weak complex.